ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Resources and Development (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** retains moisture for a long time	Its clay content holds water
2	**(d)** alluvial soil	Deposited by rivers
3	**(d)** manure and fertilisers	Leaching removes nutrients
4	**(c)** waterlogging and salinisation	Seen in Punjab and Haryana
5	**(a)** forests	Forests regenerate
6	**(c)** mining	Mining leaves deep scars
7	**(b)** a developed resource	Quality and quantity are known
8	**(c)** shelter belts	Rows of trees check wind
9	**(a)** bhangar	Older alluvium
10	**(c)** slowing the flow of water and checking soil loss	Contour ploughing decelerates water

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Arid soils lie in hot dry areas with little vegetation (1); High temperature quickens decay, so humus content remains low (1).	1+1
12	Over-irrigation raises the water table and causes waterlogging (1); Salts rise to the surface, making soil saline and infertile (1).	1+1
13	Resources are unevenly distributed across regions (1); Planning ensures balanced use and avoids depletion (1).	1+1
14	Mining removes vegetation and leaves deep scars on land (1); Dust and waste pollute the soil, causing degradation (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Technology converts materials into usable resources (1); For example, hydroelectric power turned water into energy (1); Resource creation depends on technology and culture (1).	1+1+1
16	Tree roots bind the soil (1); Leaves and branches reduce the force of rain (1); Forests add humus and slow the flow of water (1).	1+1+1
17	It must be technologically accessible (1); It must be economically feasible to use (1); It must be culturally acceptable (1).	1+1+1
	OR: Gandhi warned against greed and exploitative technology (1); Sustainable development also requires restraint in use (1); Both want resources to be conserved for future generations (1).	1+1+1
18	Some states are rich in minerals but poor in infrastructure, like Jharkhand (1); Some are rich in water but lack capital, and some have strong agriculture but few minerals (1); Unequal development results, so planning is needed (1).	1+1+1
	OR: First, resources are identified and inventoried by surveying (1); Second, a planning structure with technology and institutions is evolved (1); Third, plans are matched with national plans, e.g., for planning irrigation in a state (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Land is used for forests, farming, pasture and non-agricultural uses (1); Some is barren or wasteland (1); India has a large population on a small land share (1); This increases pressure on land (1); Planning is needed to balance uses and conserve soil (1).	1+1+1+1+1
	OR: Resources are not natural alone; human skills and technology make them (1); Water became useful for hydroelectricity with technology (1); Culture decides what is acceptable (1); Without human effort, materials remain only potential (1); So resources are functional and humans are the key (1).	1+1+1+1+1

20	Forest roots bind soil (1); Canopy reduces the impact of rain (1); Leaf litter adds humus and slows flow (1); Vegetation checks wind (1); Afforestation and shelter belts conserve soil (1).	1+1+1+1+1
	OR: Sheet erosion removes the top layer of soil over a wide area (1); Gully erosion cuts deep channels and forms ravines (1); Wind erosion removes loose soil in dry areas (1); Contour ploughing, terraces and strip cropping check water erosion (1); Shelter belts and afforestation check wind erosion (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up classifications: **renewable/non-renewable** (by exhaustibility) and **biotic/abiotic** (by origin) are different; coal is both biotic and non-renewable.
- Confusing **stock** (not yet usable for lack of technology) with **reserves** (the part of stock that can be put to use with existing technology).
- Swapping **bangar** (older alluvium, kankar nodules) and **khadar** (newer alluvium, more fertile).
- Writing that red soil is formed from alluvium; it develops on **crystalline igneous and metamorphic rocks**, and its colour is due to iron compounds.
- Calling black soil 'alluvial' or forgetting it is poor in phosphoric content though rich in lime, potash and magnesium.
- Dating the Earth Summit and **Agenda 21** at Rio de Janeiro wrongly; both are **1992**, not Kyoto or Paris.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Resources and Development (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** it contains living organisms and supports plant life	Soil is a mixture of minerals, humus and organisms
2	**(c)** black soil	Formed from volcanic rock
3	**(c)** water flows as a sheet and removes the top layer of soil	It often goes unnoticed
4	**(a)** gets fresh deposits of silt each year	New alluvium is richer in fine particles
5	**(a)** terrace farming	Steps reduce run-off on slopes
6	**(b)** Jharkhand	Many such states lack infrastructure
7	**(d)** forest soil	Forest soils vary with the mountain environment
8	**(a)** laterite soil	Typical of Kerala, Karnataka and Tamil Nadu hills
9	**(c)** using resources wisely so that future needs are also met	Meets present needs without harming the future
10	**(c)** resources are created by human use and technology	Utility depends on technology

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It is fertile with potash and lime (1); It is in level plains with water for irrigation, so crops like wheat and rice can be grown (1).	1+1
12	Heavy rain leaches the nutrients down (1); High temperature kills bacteria that break down humus (1).	1+1
13	Water flows down steep slopes with force (1); In clayey soils it cuts deep channels (1).	1+1
14	It is made of fine clay particles, which absorb water and swell (1); In dry weather the clay shrinks and deep cracks appear (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Heavy rain leaches nutrients from laterite (1); Humus is low because bacteria cannot survive the heat (1); So manures and fertilisers are needed for crops (1).	1+1+1
16	Different soils suit different crops (1); It allows varied agriculture in different regions (1); It helps in planning land use and conservation (1).	1+1+1
17	Too many animals graze on a limited area (1); Grass cover is lost and the soil is exposed (1); Wind and water remove the soil, leading to degradation, as in Gujarat and Rajasthan (1).	1+1+1
	OR: Contour ploughing is done along the slope contour (1); Strip cropping grows crops in strips with grass in between (1); Both slow run-off, but strip cropping also checks wind (1).	1+1+1
18	Community resources are owned by all members, e.g., village pastures (1); Rules are set by the community for their use (1); Everyone has a right to use them but must conserve them (1).	1+1+1
	OR: Soil contains living organisms (1); It supports plant and animal life (1); It takes thousands of years to form, so it needs care (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Alluvial soil covers the northern plains and deltas (1); Black soil covers the Deccan lava plateau (1); Red and yellow soils are found in the south and east (1); Laterite soils are found in the Western Ghats and hill areas (1); Arid soils are in Rajasthan and forest soils in hills (1).	1+1+1+1+1
	OR: Sustainable development means meeting needs without harming the future (1); The Rio Summit adopted Agenda 21 in 1992 (1); Gandhi said that there is enough for need but not for greed (1); Both oppose over-exploitation (1); Local planning and conservation make it possible (1).	1+1+1+1+1

20	Mining degrades land in Jharkhand, Chhattisgarh and Odisha (1); Overgrazing damages land in Gujarat and Rajasthan (1); Over-irrigation causes salinity in Punjab and Haryana (1); Remedies are afforestation, shelter belts and control of grazing (1); Proper disposal of industrial waste also helps (1).	1+1+1+1+1
	OR: Resources include biotic and abiotic, renewable and non-renewable (1); Some resources like fossil fuels are limited (1); Over-exploitation causes depletion (1); Conservation is needed to maintain supply for future (1); Planned use meets present and future needs (1).	1+1+1+1+1

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- Confusing **stock** (not yet usable for lack of technology) with **reserves** (the part of stock that can be put to use with existing technology).
- Swapping **bangar** (older alluvium, kankar nodules) and **khadar** (newer alluvium, more fertile).
- Writing that red soil is formed from alluvium; it develops on **crystalline igneous and metamorphic rocks**, and its colour is due to iron compounds.
- Calling black soil 'alluvial' or forgetting it is poor in phosphoric content though rich in lime, potash and magnesium.
- Dating the Earth Summit and **Agenda 21** at Rio de Janeiro wrongly; both are **1992**, not Kyoto or Paris.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Resources and Development (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** growing crops in alternate strips with grass to break wind and water flow	Grass strips check run-off
2	**(b)** village pasture	Common to all members of the community
3	**(d)** resources are unevenly distributed	Regions differ in resource endowment
4	**(d)** deep channels cut the fields and create ravines	Chambal ravines are an example
5	**(c)** arid soil	Dry climate and sandy texture
6	**(d)** land not available for cultivation	Built-up land is non-agricultural use
7	**(b)** Punjab	Fertile alluvial plains
8	**(a)** the greed of man and exploitative modern technology	Greed leads to over-exploitation
9	**(c)** great pressure on land resources	Land per person is small
10	**(a)** a potential resource	Example: wind energy in Rajasthan and Gujarat

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Stock cannot be used because of lack of technology (1); Reserves can be used with the present technology but their use has not started (1).	1+1
12	They reduce wind speed and protect the soil (1); They stabilise sand dunes and shield crops (1).	1+1
13	Potential resources are found in a region but not yet used (1); Developed resources have been surveyed and are in use (1).	1+1
14	Resources do not exist by themselves; they are created by human activities (1); A thing becomes useful only with technology and culture (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	International institutions regulate oceanic resources beyond 200 nautical miles (1); Summits like Rio 1992 set global agendas such as Agenda 21 (1); They guide countries to cooperate in sustainable use (1).	1+1+1
16	Khadar is new alluvium, finer and more fertile (1); Bhangar is old alluvium with kankar nodules and is less fertile (1); Khadar is renewed by floods, while bhangar is on higher ground (1).	1+1+1
17	Potential resources are not yet used (1); Developed resources are surveyed and in use (1); Stock and reserves are the part not used because of technology or other limits (1).	1+1+1
	OR: Black soil is clayey, forms from lava and retains moisture (1); Red soil forms on crystalline rocks and has a lower water-holding capacity (1); Black soil suits cotton, while red soil is used for millets and groundnut after fertilisers (1).	1+1+1
18	Planting trees stabilises the soil (1); Roots hold soil and canopy reduces rain impact (1); It prevents erosion and gullies on slopes (1).	1+1+1
	OR: Cities and roads convert farmland into built-up land (1); Net sown area shrinks (1); Pressure on the remaining land increases (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	First, identification and inventory of resources (1); Second, a planning structure with technology and skill is evolved (1); Third, plans are matched to national development plans (1); Planning is done at national, state and local levels (1); Aim is balanced and sustainable development (1).	1+1+1+1+1
	OR: Individual resources are private, e.g., plots and houses (1); Community resources are shared, e.g., pastures and parks (1); National resources lie within political boundaries (1); International resources are in the oceans beyond 200 nautical miles (1); Each has different rules of use (1).	1+1+1+1+1

20	Alluvial soil is deposited by rivers in the plains (1); Black soil is formed from lava rocks (1); Alluvial soil is rich in potash and lime; black soil is rich in clay and retains moisture (1); Alluvial suits wheat, rice and sugarcane (1); Black soil suits cotton (1).	1+1+1+1+1
	OR: Jharkhand and Chhattisgarh are rich in minerals but lack development (1); Rajasthan has solar and wind potential but little water (1); Punjab has fertile land but few minerals (1); Colonial exploitation affected resource development in some areas (1); Planning is needed to balance resources and development (1).	1+1+1+1+1

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